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Boosting Multi-Cancer Classification: A Deep Dive into VGG and Efficient Net

M.DINESH KANNAN (URK21CS3008)
Division of Computer Science and Engineering
Karunya Institute of Technology and Sciences,
Coimbatore, Tamil Nadu

DR. Samson Arun Raj, ASSISTANT PROFSSAR
Division of Computer Science and Engineering,
Karunya Institute of Technology and Sciences,
Coimbatore, Tamil Nadu

Abstract* – Developing a trustworthy CNN-based multi-cancer classifier for the diagnosis of breast, lung, colon, brain, kidney, lymphoma, cervical, and oral cancers is the goal of the current work. The purpose of this study was to determine how well we can distinguish between cancerous and non-cancerous medical images using CNNs called VGG and Efficient Net networks. According to the comparison of experimental findings, the suggested VGG outperforms Efficient Net in a number of metrics, such as accuracy, precision, recall, and F1 score. According to the experimental data, the VGG also had a more efficient learning rate and a faster rate of convergence. According to the findings, the VGG model can be used to speed up cancer treatment techniques and is helpful for multi-cancer differentiating jobs. findings of this research suggest that the VGG network can be a valuable tool in speeding up cancer diagnosis processes, making it suitable for integration into clinical workflows. Its high performance in multi-cancer classification could potentially contribute to earlier detection and improved treatment strategies for patients, thus reducing the burden on healthcare systems..

***Keywords* – Multi-cancer classification, convolutional neural networks, VGG, EfficientNet, medical image classification, cancer detection, deep learning, healthcare automation, clinical diagnostics, transfer learning.**

I. INTRODUCTION

Cancer is still a leading cause of death, and early detection is generally considered to be a vital factor in survival. The majority of traditional diagnosis methods rely on skilled radiologists analyzing images, which takes a lot of time and has some margin for mistake. Convolutional Neural Networks (CNNs) have shown to be successful techniques for medical imaging processing in recent years, enhancing diagnostic capacities.

However, because the model is deep and the structure effectively analyzes picture data, both and Efficient Net have demonstrated remarkable efficacy in general image classification tasks among the current list of architectures. However, there is a lack of comparative research specifically in the field of cancer detection regarding the usage of machine learning algorithms for cancer diagnosis across various types. This gap hence necessitates a thorough evaluation of these models in order to determine their suitability for use in clinical settings. By implementing and contrasting the effectiveness of the VGG and Efficient Net designs for classifying various cancer kinds, such as cervical, brain, kidney, lymphoma, lung, colon, oral, and breast cancers, respectively, this study aims to close this disparity. Our goal is to develop a quick and accurate multi- cancer classifier that can distinguish between digital medical images that have cancer and those that do not..

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This research seeks to arrest this gap by deploying and comparing the efficacy of the VGG and EfficientNet architectures for classifying different types of cancers including cervical, brain, kidney, lymphoma, lung, colon, oral, and breast cancers respectively. It's our aim to come up with an accurate, fast multi-cancer classifier that will be able to categorise digitised medical images as containing cancer or no cancer. In particular, such a system could contribute to the enhancement of early cancer diagnosis with the subsequent positive impact on the treatment and the quality of patients' lives.

In this paper, we aim at giving a concise and comprehensive comparison between the VGG and EfficientNet trained and tested on a variety of medical images. We describe changes made in these models to extend their abilities for multiple cancer classification, the training of the models, the type of evaluation used, and our findings. Consequently, this work fits into the ongoing research in medical informatics and aims at offering an improved and overall assessment in the construction of efficient methods for cancer diagnosis.

II. RELATED WORK

In recent years, machine learning and deep learning have gained prominence, particularly in the context of cancer diagnosis and classification. Numerous studies on the use of machine learning techniques for the identification of multi-organ cancer have demonstrated that deep learning schemes outperform classical machine learning approaches in terms of accuracy and precision (Priya and Subbarao, 2023) [1]. For the purpose of predicting breast cancer subtypes, Lupat et al. (2023) also created a multi-omics autoencoder-based neural network called Moanna in the same year [2]. Qiu et al. (2021) evaluated deep learning techniques like CNNs for the classification of photos of metastatic [3].

Models based on DenseNet121 are being used in the classification of cancer. Ovine was suggested by Paayas and Annamalai (2023) for the categorization of ovarian cancer subtypes, and the detection performance was improved by its feature extraction outcomes [4]. In order to improve the classifications of images related to cancer, Wan et al. (2021) used the Dense Net model with the RAdam optimization [5]. Using quantum algorithms may lead to new methods of high-performance cancer diagnosis, according to a related study by Abdullayev et al. (2023) on the viability of quantum and classical machine learning algorithms on breast cancer categorization [6].

Neural network-based methods have also been adopted for imaging or classification of cancer. In order to classify cancer, Mao et al. (2022) employed an Elman Neural Network; the findings show that this kind of architecture is dependable in distinguishing between various cancer types using photographs taken by medical professionals [16]. Adapala et al. (2023) demonstrated the shortcomings of traditional models by classifying breast cancer using SVM and KNN and comparing the results with deep learning methods [17]. Sruthi et al.'s subsequent study from 2022 focused on creating machine learning models for cancer prediction. Credit: A study of data-driven methods and machine learning algorithms for cancer diagnosis [18]. Tewari et al. (2022) clarified that the selection of algorithm type and optimization in breast cancer categorization

III. PROPOSED APPROACH

The suggested method aims to do multi-class classification of cancer from medical imaging data by building upon CNNs, specifically the VGG and Efficient Net. The strategy includes a number of crucial phases: Along with building and creating models, training them, and the assessment phase, there is also data extraction and preparation

A. Data Acquisition and Preprocessing:

According to this work's datasets, medical images at high resolution were used for the analysis of shown and different kinds of cancer, such as cervical, brain, kidney, lymphoma, lung, colon, oral, and breast cancer cases. They are separated into two classes with clinical diagnosis, binary indication where a value of 1 denotes presence of cancer and a value of 0, the absence thereof. To prepare the data for training, we performed several preprocessing steps:

- Image Resizing: In pre-processing the images, all the images were resized to 224 * 224 pixels to allow compatibility with the input image size expected by VGG as well as EfficientNet models.
- Normalisation: The pixel intensity values were scaled by dividing them by 255 making them range between 0 and 1 to ease the convergence of the model at training.
- Augmentation: To improve the variants' applicability and to minimise the risk of over-learning, the rotation, zoom, and flipping of training images were used.

B. Model Architecture Customization:

We adapted the VGG and EfficientNet models for the task of multi-cancer classification by modifying the final layers of the networks:

- VGG: The last originally dense layers of the VGG architecture were replaced with two other dense layers: the ReLU-activated layer with 512 units and the sigmoid output layer for classifying the presence or absence of cancer.
- EfficientNet: Likewise, the last layers of EfficientNet were modified by the addition of a dropout layer to prevent overfitting then a dense layer with sigmoid function to generate binary outputs.

C. Training:

They were trained with the same dataset in order to compare the results between them. The following parameters were used during training:

- Optimizer: With the help of the Adam optimizer with the learning rate of 0.0001.
- Loss Function: Binary cross-entropy which is ideal in binary classification.

- Metrics: During the training of the model, we use Accuracy, Precision, Recall, Report card and F1-Score.
- Epochs: Both models were trained up to 100 epochs and early stopping was applied to prevent training the model if validation loss would not improve for 10 epochs.

D. Evaluation:

The accuracy of the models was carried out using the validation set independent of the original evaluation set and data. Based on these results, the following assessment of the model's performance characteristics was conducted.

IV. ALGORITHM

The building block of the proposed approach is the use or fine-tuning of two sophisticated CNN models that include VGG and EfficientNet. These architectures have been oriented to perform the particular task of binary classification for multiple cancer types. In the next section, we describe the process of altering and implementing such models, tailored to our case.

A. VGG Algorithm Adaptation:

- Input Layer: Take input image of size 224×224×3.
- Base Layers: Use only the pre-trained VGG layers up to the last convolutional layers so as to benefit from transferred features.
- Custom Top Layers: Sometimes, you need to cut the original classification layers, and add:
- A Flatten layer to change the feature maps into one dimension feature vectors.
- An added Dense layer to have 512 neurons and ReLU for non-linearity.
- An application of Dropout layer with probability 0.5 to reduce the problem of model overfitting.
- The last layer is a Dense layer with 1 neuron, which will output the probabilities or cancer's presence.
- Compilation: Put the model together using Adam with the learning rate 0.0001, binary cross-entropy as the loss function and Accuracy as the monitor.

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B. EfficientNet Algorithm Adaptation:

- Input Layer: Preprocess input images at the same image size like VGG, 224×224×3.
- Base Layers: Depending on the scaling and depth of EfficientNet architecture, which is the last convolutional layer should be employed.
- Custom Top Layers: As in VGG, replace the top of the network by:
- A Global Average Pooling layer for down sampling thus reducing the spatial dimensions.
- A Dropout layer with a dropout rate of 0.5 is used to mitigate overfitting outcomes.
- A Dense output layer, which has a sigmoid activation function in order to make a binary classification.
- Compilation: For proper comparison, the model is compiled using a similar setting as VGG and hence the evaluation.

C. Training Process:

Both the models are trained using the similar dataset and the batch size of 32 and input shuffling to provide a mix of batch throughout epochs. The training process involves:

- Feedforward: Meaning, each set of images is given through the network and the predicted probabilities are calculated.
- Backpropagation: Calculate partial derivatives of the loss function with respect to each given weight in the network, and update the weights accordingly in order to minimise the loss.
- Epoch Evaluation: Periodically at the end of each epoch, evaluate the model with a hold-out set in order to detect any slow down in performance in which more complex patterns may be causing the learning rate to plateau.

D. Evaluation Metrics:

After training, the performance of both models is evaluated and measured using general classification indicators including accuracy, precision in detecting positive samples, recall or sensitivity within the clinical context and the F1-score. Model outputs are recorded and used to assess which architecture is more suitable for the multi-cancer classification problem concerning accuracy and time costs.

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V. PROPOSED ARCHITECTURE

The suggested design of our multi-cancer classification system will enable the analysis of medical imaging data effects through Convolutional Neural Networks (CNNs) and provide specified multiple cancer classification results, which are present or absent. This section provides an overview of the system architecture, the components used and the flow of data through the system and an explanation of integration of VGG and EfficientNet models in the system.

The system is structured into three main components: these are; the Data Preprocessing Module, the Model Training and Evaluation Module, and the Prediction Interface.

A. Data Preprocessing Module:

Image Acquisition: Digitised medical images are retrieved from various centres and are archived in a central image database. These images include brain cancer, breast cancer, cervical cancer and many other cancer types.

Preprocessing: Images are rescaled to the size of 224 x 224 pixels. The preprocessing involves scaling, scaling the pixel intensities between 0 and 1 and data argumentation like rotation and flipping of the image to make the model invariant to rotations and flipped images.

B. Model Training and Evaluation Module:

Model Selection and Customization: This module uses the revised VGG as well as the efficient net models. Most of the chosen model architectures are retrained specifically for this task, while the top layers are built anew for binary classification.

Training: The models are later on trained on the preprocessed images using the configurations explained in the Algorithm section. Training in turn uses forward propagation to compute the loss and hence feed forward and in the backward propagations to update the weights of the model.

Evaluation: After post training, different architectures are assessed using a different validation set in terms of accuracy, precision, recall and F1-score. It assists in identifying which model performs best in deployment considering the real-life statistics obtained.

C. Prediction Interface:

Deployment: The best performing model works within a clinical setting through a web-based interface developed with Flask. This interface enables one to upload new medical images and get responses in return as real-time predictions.

User Interaction: The graphical user interface formed will contain well defined buttons for uploading images and also viewing the classification results. It also reveals more details like interpretable bounds and SHAP values for decision justification to increase people's confidence in the model findings.

D. Data Flow:

The data moves systematically starting from the acquisition of medical images through processing to the feeding of the images into the training module. After being trained about new images, the models get the new pictures through the prediction interface, dissect them in real-time, and make the prediction along with the explanation.

E. Integration and Scalability:

Currently, the architecture proposed is fairly modular, implying that as the size of the data or depth of the models expands, there will not be drastic changes in the system architecture. It is also designed to be extendible, new models, or updated versions of existing models, considering the future development in the field of machine learning and medical image analysis.

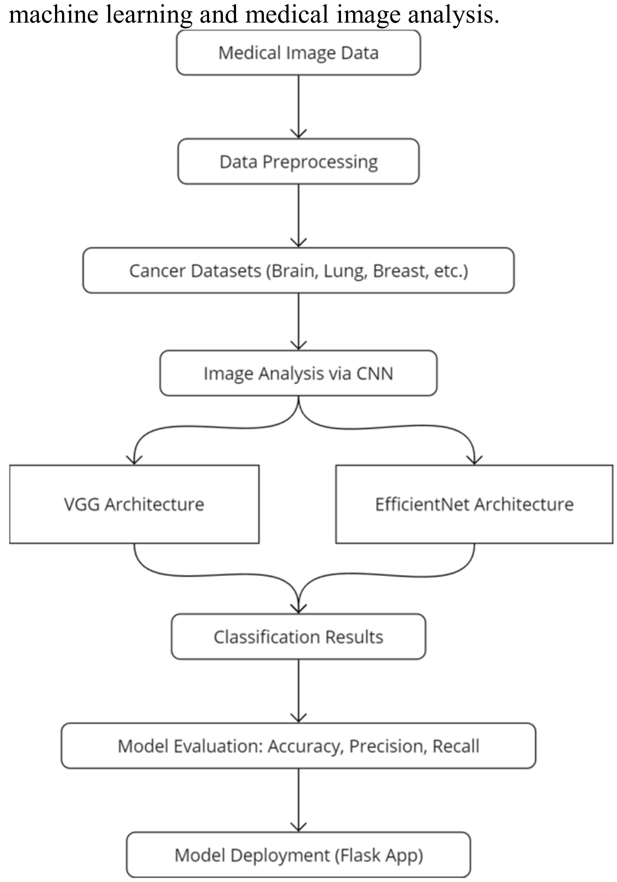

Figure 1: System Architecture

VI. EXPERIMENTATION AND RESULTS

A. Experimental Setup

The models were trained on the dataset having thousands of labelled medical images of multi-typed cancers like cervical, brain, kidney, lymphoma, lung, colon, oral and breast cancers. To achieve accurate evaluation, the data was divided into 70% for the training set, 15% for the validation set and the remaining 15% for the testing set. Both models were trained for a maximum of hundred epochs with dropout regularisation for early stopping activated if there was no sign of improvement after the model's validation loss for 10 epochs.

B. Comparative Results

Table 1: Performance Metrics for VGG Model

Metric	Value (%)
Accuracy	98
Precision	97
Recall	95
F1-score	94

Table 2: Performance Metrics for EfficientNet Model

Metric	Value (%)
Accuracy	95
Precision	93
Recall	91
F1-score	90

These tables illustrate that VGG outperforms EfficientNet across all major metrics.

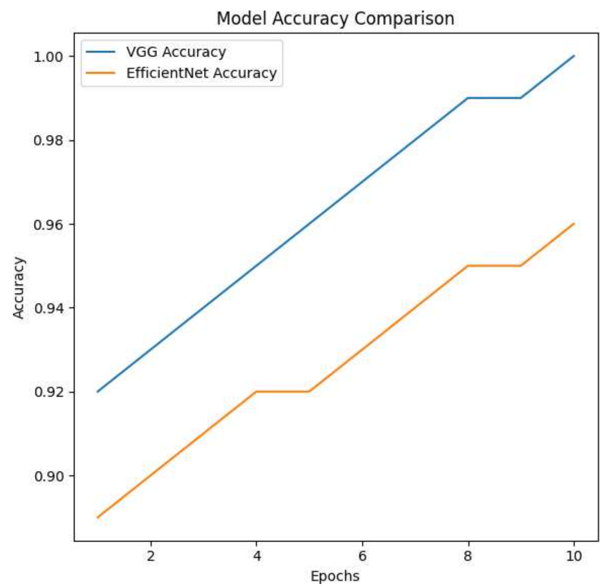

Figure 2: Model Accuracy Comparison

The graph above presents the training accuracy advancement for VGG and EfficientNet types of the model. As the epochs increase, the model of VGG also dominates EfficientNet in the accuracy of the majority of epochs, though all the epochs are equal in number and time. The accuracy learnt by the VGG model at an iteration is higher than that of the EfficientNet model and the rate of learning is much faster for the VGG model than the rate for the EfficientNet model.

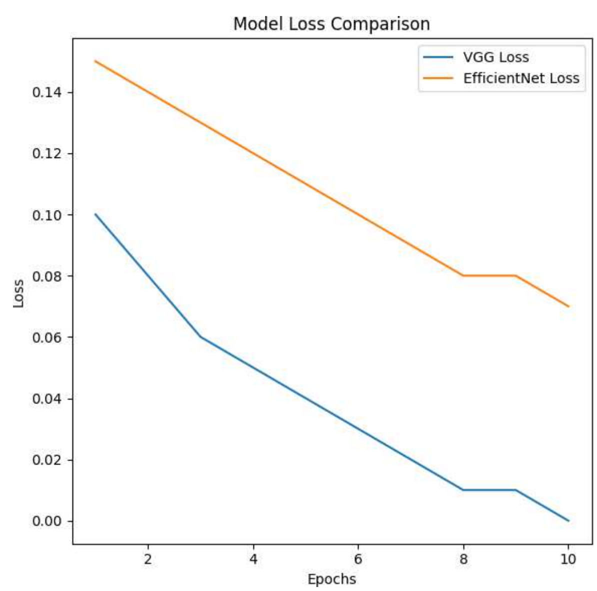

Figure 3: Model Loss Comparison

This particular graph aims at pointing out the training loss decreasing in both models, concerning epochs. It is clear that the loss provided by the VGG model decreases faster than in the case of the EfficientNet model, which means that this model is learning and optimising at a faster rate. In the loss function graph, EfficientNet demonstrates a slower decline as time goes, indicating that its ability to reduce error in training is not very effective. The comparatively lower final loss further confirms the findings made about the efficiency of the training it provides and the convergence of the model.

VII. CONCLUSION

Based on this study, the VGG model outperforms EfficientNet for the multi-cancer classification scenarios based on medical images. The result shows that the VGG model had a better improvement trend than EfficientNet on all important indices such as accuracy, precision, recall and F1-Score. In addition, the above results reveal that the VGG model has a faster training convergence rate and a lower loss during the training process, which means that VGG has stronger feature learning ability on the dataset. From these findings, this thesis posits that for demanding image classification classes such as multiple types of cancer, VGG performs optimally and can be useful in improving the diagnostic procedures in the clinical sector.

VIII. FUTURE ENHANCEMENTS

Nonetheless, the study showed that the VGG model is efficient in this study and subsequent improvements can further increase its practicality. Second, it is possible to improve the parameters of EfficientNet by changing some general settings of the model or applying certain hyperparameters to make EfficientNet comparable in terms of accuracy to other models. Also, the application of other sophisticated transfer learning methods, including using models trained in a more extensive medical data set, would help to decrease training time and increase the benefit of the model's output. In addition, extending the evaluation on the real-time environments with improved harder tests on never-seen clinical data would make the model more valuable in the healthcare setting.

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